

STUDENT NUMBER: _____

TEACHER: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

HALF YEARLY EXAMINATION

2010

MATHEMATICS

EXTENSION 1

GENERAL INSTRUCTIONS:

- Attempt **ALL** questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet.
- Marks indicated for each question are only a guide and could change.

QUESTION 1**Marks**

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x}$ **1**
- (b) (i) Sketch the graph of $y = -x(x + 2)(x - 3)$ without using calculus. **2**
- (ii) Solve $\frac{6}{x} > x - 1$ **3**
- (c) If A and B are the points $(2, -1)$ and $(-3, 5)$ respectively, find the co-ordinates of the point $P(x, y)$ that divides the interval AB externally in the ratio 3:4. **2**
- (d) (i) Show that the curves $y = \sin 2x$ and $y = \cos 2x$ intersect at $x = \frac{\pi}{8}$. **1**
- (ii) Find the acute angle between the two curves at $x = \frac{\pi}{8}$. **3**

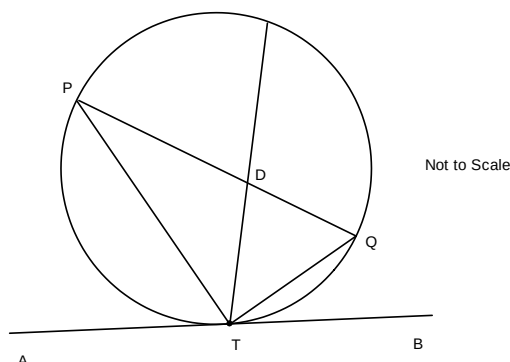
QUESTION 2 (Start a new page)

- (a) Find the exact value of $\cos 15^\circ$ **2**
- (b) If α, β and δ are the roots of the cubic $2x^3 + 6x^2 - 4x + 5 = 0$, find :
- (i) $\alpha + \beta + \delta$. **1**
- (ii) $\alpha\beta + \alpha\delta + \beta\delta$. **1**
- (iii) $\alpha^2 + \beta^2 + \delta^2$. **2**
- (c) Differentiate $\log_e \sqrt{\frac{x+1}{x-1}}$. **2**
- (d) (i) Express $\sin x + \sqrt{3}\cos x$ in the form $A\sin(x + \alpha)$. **2**
- (ii) Hence solve $\sin x + \sqrt{3}\cos x = 1$ for $0 \leq x \leq 2\pi$. **2**

QUESTION 3 (Start a new page)

Marks

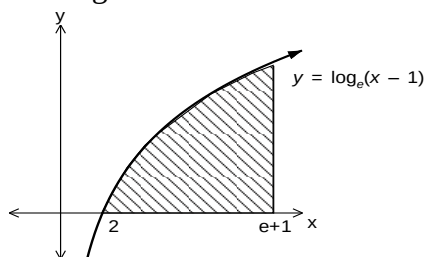
- (a) Solve $2 \log_e(x + 2) = \log_e(5x + 6)$ 2
- (b) Evaluate $\int_0^{\frac{\pi}{3}} 2 \cos^2 x \, dx$. 3
- (c) Taking $x = 2.5$ as the first estimate for the root of $f(x) = \sin x - \ln x$, use one application of Newton's method to find a better estimate for the root to 3 decimal places. 3
- (d) In the diagram below ATB is a tangent and PT and QT bisect the angles $\angle ATD$ and $\angle DTB$ respectively.



- (i) Redraw this diagram on your page then prove that PQ is the diameter of the circle. 2
- (ii) Prove $PQ \perp DT$. 2

QUESTION 4

- (a) The rate at which a body cools is proportional to the difference between the temperature (T) of the body and the surrounding temperature (C).
ie. $\frac{dT}{dt} = k(T - C)$
- (i) Prove that $T = C + Ae^{kt}$ is a solution to the differential equation above. 1
- (ii) A heated body cools from 100°C to 60°C in an hour, after being placed in a room with a temperature of 20°C . Find the temperature of the body after a further 2 hours. 3
- (b) Find the area of the shaded region below: 4



- (c) (i) Prove $2^2 + 2^3 + 2^4 + \dots + 2^{n+1} = 2^2(2^n - 1)$ by mathematical induction. 3
- (ii) Hence evaluate $2^2 + 2^3 + \dots + 2^{18}$. 1

QUESTION 5 (Start a new page)

Marks

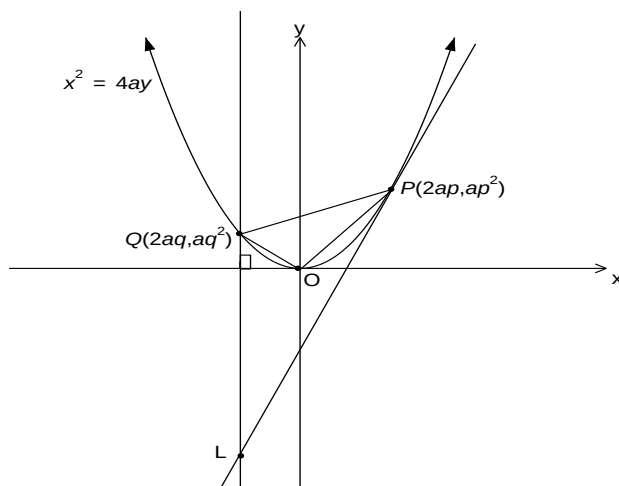
- (a) (i) Sketch on the same set of axes the graphs of $y = 2x + 1$ and $y = |x - 2|$

2

- (ii) Hence or otherwise solve $|x - 2| < 2x + 1$

2

- (b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.



- (i) Find the gradient of OP .
- (ii) The chord PQ subtends a right angle at the origin. Show that $pq = -4$.
- (iii) Show that the equation of the tangent at P is $y = px - ap^2$.
- (iv) The tangent at P meets the line through Q perpendicular to the x axis at L . Show that L has co-ordinates $(2aq, 2apq - ap^2)$.
- (v) Find the locus of L .

1

2

2

1

2

QUESTION 6 (Start a new page)

- (a) Evaluate $\int_0^{\frac{2\pi}{3}} \sec^2 x \tan^2 x \, dx$

3

- (b) If $\log_a 3 = x$ and $\log_a 4 = y$ express $\log_3 6$ in terms of x and y .

2

- (c) Hayden invests \$2000 each year in a superannuation fund which earns 5% compound interest per annum for n years.

- (i) How much does his first investment amount to after n years?
- (ii) Show that the total of his investments after n years is $42000(1.05^n - 1)$
- (iii) Find the value of n if the total of his investments after n years is \$93 454.20.
- (iv) $y = ax^3 - 7x^2 + bx + 20$ has a double root at $x = 2$. Find the values of a and b .

1

2

1

3

QUESTION 7 (Start a new page)

Marks

(a) Prove

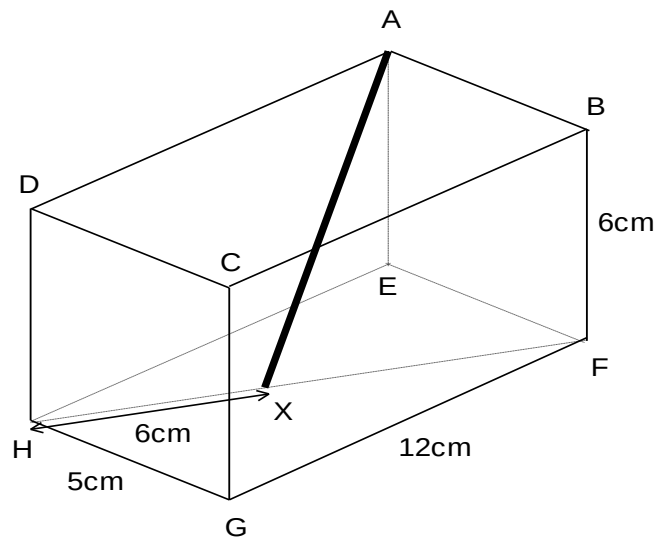
$$\frac{\sin^3 a + \cos^3 a}{\sin^2 a - \cos^2 a} = \frac{\operatorname{cosec} a + \cot a}{1 + \cot a}$$

3

(b) If $\frac{9^x + 6^x}{15^x + 10^x} = a^x$ find a .

2

(c)



(i) An open pencil case in the shape of a rectangular prism has dimensions 5 cm by 6 cm by 12 cm . A pencil AX is placed in the case such that it rests at points A and X where X is 6 cm along the diagonal FH . What angle does the pencil AX make with the base (plane $EFGH$)?

3

(ii) A lid is placed on the pencil case. An ant stands at point D . It walks on the outside of the pencil case to F . What is the shortest distance from D to F ?

1

(d) (i) A function $y = f(x)$ has the following properties :

$$\begin{aligned} y' &= \frac{1}{2}y \\ y'' &= \frac{1}{2}y' \\ y''' &= \frac{1}{2}y'' \quad \text{etc.} \end{aligned}$$

Give a possible equation for $y = f(x)$.

1

(ii) Find $\lim_{n \rightarrow \infty} (y' + y'' + y''' + \cdots + y^n)$

2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

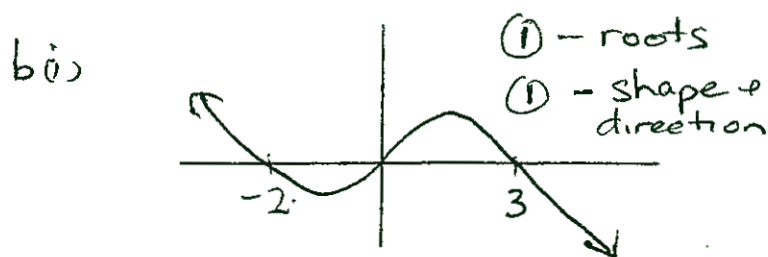
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

Solutions.

a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \frac{2}{3}$ ①



(ii) $\frac{6}{x} > x-1$

$6x - x^2(x-1) > 0$ ①

$x(6 - x(x-1)) > 0$

$x(6 - x^2 + x) > 0$

$-x(x^2 - x - 6) > 0$ } ①

$-x(x+2)(x-3) > 0$

$x < -2$ or $0 < x < 3$ ①

c) $(2, -1)$ $(-3, 5)$ ① mark for $(-\frac{1}{7}, \frac{11}{7})$

$-3 : 4$

① $(\frac{-3(-3) + 4(2)}{-3+4}, \frac{-3(5) + 4(-1)}{-3+4})$ ①
 $= (17, -19)$

d) $\sin 2x = \cos 2x$

$\tan 2x = 1$

$2x = \frac{\pi}{4}, \frac{5\pi}{4}, \dots$

$x = \frac{\pi}{8}, \dots$ ①

$m_1 = 2 \cos 2(\frac{\pi}{8})$ $m_2 = -2 \sin 2(\frac{\pi}{8})$

$m_1 = \frac{2}{\sqrt{2}}$ $m_2 = -\frac{2}{\sqrt{2}}$ ①

① $\tan \theta = \left| \frac{\frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}}}{1 + \frac{2}{\sqrt{2}} \cdot \frac{2}{\sqrt{2}}} \right| = \left| \frac{\frac{4}{\sqrt{2}}}{-1} \right|$

$\theta = 70^\circ 32'$ ①

Question 2

a) $\cos 15^\circ = \cos(45^\circ - 30^\circ)$

① $= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$

$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}$ or $\frac{\sqrt{6}+\sqrt{2}}{4}$ ①

b) $2x^3 + 6x^2 - 4x + 5 = 0$

(i) $\alpha + \beta + \gamma = -\frac{6}{2}$ (ii) $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{4}{2}$

① $= -3$ ① $= -2$

(iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ ①
 $= (-3)^2 - 2(-2)$
 $= 13$ ①

c) $y = \log_e \sqrt{\frac{x+1}{x-1}}$

$= \frac{1}{2} [\log(x+1) - \log(x-1)]$ ①

$\frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{x+1} - \frac{1}{x-1} \right]$ ①
 $= \frac{-1}{(x+1)(x-1)}$

d) $A \sin x \cos x + A \cos x \sin x = \sin x + \sqrt{3} \cos x$

$A \cos x = 1$ $A \sin x = \sqrt{3}$

$A = \sqrt{(\sqrt{3})^2 + 1^2}$ $\tan x = \sqrt{3}$

$= 2$ ① $x = \frac{\pi}{3}$ ①

e) $2 \sin(x + \frac{\pi}{3}) = 1$

$\sin(x + \frac{\pi}{3}) = \frac{1}{2}$

$\therefore x + \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$ ①

$x = -\frac{\pi}{6}, \frac{\pi}{2}$

$\therefore x = \frac{11\pi}{6}, \frac{\pi}{2}$ ①

Question 3.

a) $2 \log_e(x+2) = \log_e(5x+6)$

$$\log_e(x+2)^2 = \log_e(5x+6)$$

$$\therefore (x+2)^2 = 5x+6 \quad (1)$$

$$x^2 + 4x + 4 - 5x - 6 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$(1) \quad x = 2, -1 \quad (x > -2 \quad \checkmark)$$

b) $\int_0^{\frac{\pi}{3}} 2 \cos^2 x \, dx$

$$= \int_0^{\frac{\pi}{3}} (\cos 2x + 1) \, dx \quad (1)$$

$$= \left[\frac{1}{2} (\sin 2x) + x \right]_0^{\frac{\pi}{3}}$$

$$= \left(\frac{1}{2} \sin\left(\frac{2\pi}{3}\right) + \frac{\pi}{3} \right) - (0+0)$$

$$= \frac{\sqrt{3}}{4} + \frac{\pi}{3}$$

c) $x_1 = 2.5 - \frac{f(2.5)}{f'(2.5)}$

$$f(2.5) = \sin(2.5) - \ln(2.5)$$

$$= -0.3178 \dots \quad (1)$$

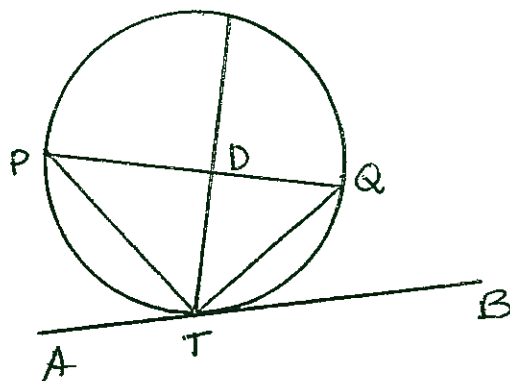
$$f'(x) = \cos x - \frac{1}{x}$$

$$f'(2.5) = \cos 2.5 - \frac{1}{2.5} = -1.20 \dots \quad (1)$$

$$x_1 = 2.5 - \frac{-0.3178}{-1.20}$$

$$= 2.235 \quad (1)$$

doi)



let $\angle ATP = x^\circ$

$\therefore \angle PTD = x^\circ$ (Given PT bisects $\angle ATD$)

let $\angle DTQ = y^\circ$

$\therefore \angle QTB = y^\circ$ (Given QT bisects $\angle DTB$)

$\therefore 2x + 2y = 180^\circ$ (straight angle) (1)

hence $x + y = 90^\circ$

$\therefore \angle PTQ = 90^\circ$

If $\angle PTQ = 90^\circ$ then PQT must
(1) be in a semi circle
 $\therefore$ PA is the diameter.

(ii)

$\angle ATP = \angle PQT = x^\circ \quad (1)$

(Angle between tangent and chord
= angle in the alternate segment)

$\therefore \angle TDQ = 180 - (x + y)$ (Angle Sum of Δ)

(1) $\therefore \angle TDQ = 90^\circ$

$\therefore DT \perp PQ$

Question 4

Q (i) $T = C + Ae^{kt} \rightarrow T - C = Ae^{kt}$

$$\frac{dT}{dt} = k(T - C)$$

$$\left. \begin{aligned} \frac{dT}{dt} &= kAe^{kt} \\ &= k(T - C) \end{aligned} \right\} \textcircled{1}$$

$\therefore T = C + Ae^{kt}$ satisfies $\frac{dT}{dt} = k(T - C)$

(ii) $C = 20$

$T = 20 + Ae^{kt}$ when $t=0, T=100$

$\therefore 100 = 20 + Ae^0$

$A = 80$

$\therefore T = 20 + 80e^{kt}$ $\textcircled{1}$

when $t=1, T=60$

$\therefore 60 = 20 + 80e^k$

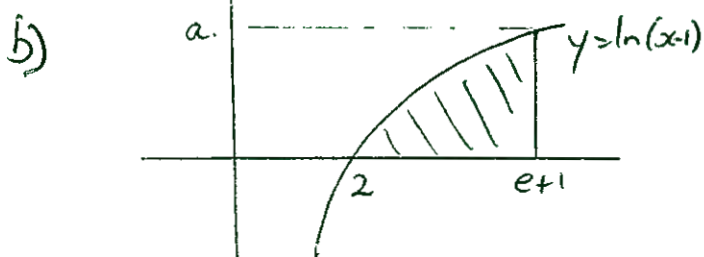
$\frac{1}{2} = e^k$

$\textcircled{1} \begin{cases} \ln\left(\frac{1}{2}\right) = k \ln e \\ k = -0.693 \dots \end{cases}$

when $t=3$

$T = 20 + 80e^{-0.693 \dots (3)}$

$T = 30$ $\textcircled{1}$



Area = $\square - \text{shaded area}$

$a = \ln(e+1-1)$
 $= 1$

$\therefore \text{Rectangle} = (e+1) \times 1$
 $= e+1$ $\textcircled{1}$

$y = \log_e(x-1) \therefore x-1 = e^y$
 $x = e^y + 1$

Area = $\int_0^1 (e^y + 1) dy$ $\textcircled{1}$

$= [e^y + y]_0^1$ $\textcircled{1}$

$= (e+1) - (e^0 + 0)$

$= e$

$\therefore \text{Required area} = e+1 - e$
 $= 1$ $\textcircled{1}$

c) Prove $2^2 + 2^3 + \dots + 2^{n+1} = 2^2(2^n - 1)$

Step 1. Prove true for $n=1$

i.e. $2^2 = 2^2(2^1 - 1)$

$4 = 4$ $\checkmark$

Step 2 Assume true for $n=k$.

i.e. $2^2 + \dots + 2^{k+1} = 2^2(2^k - 1)$ $\textcircled{1}$

Step 3. Prove true for $n=k+1$

i.e. $2^2 + \dots + 2^{k+1} + 2^{k+2} = 2^2(2^{k+1} - 1)$

by assumption need to prove

$\Rightarrow 2^2(2^k - 1) + 2^{k+2} = 2^2(2^{k+1} - 1)$ $\textcircled{1}$

LHS $2^{k+2} - 2^2 + 2^{k+2}$

$= 2(2^{k+2}) - 2^2$

$= 2^{k+3} - 2^2$ $\textcircled{1}$

$= 2^2(2^{k+1} - 1)$

$= \text{RHS.}$

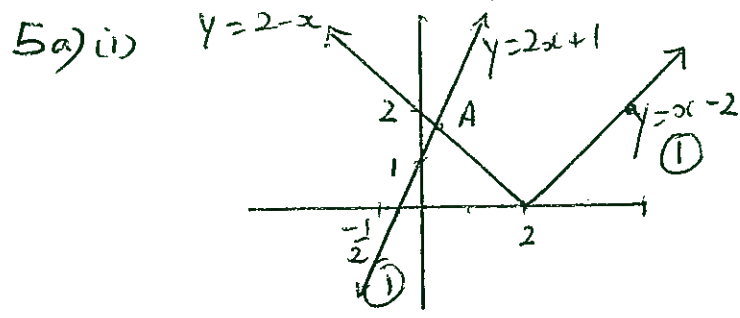
Step 4 Proved true for $n=1$ &

assumed true for $n=k$ & prove

true for $n=k+1 \therefore$ true for

$n=1, n=2, \dots$ & for all n

by mathematical induction.

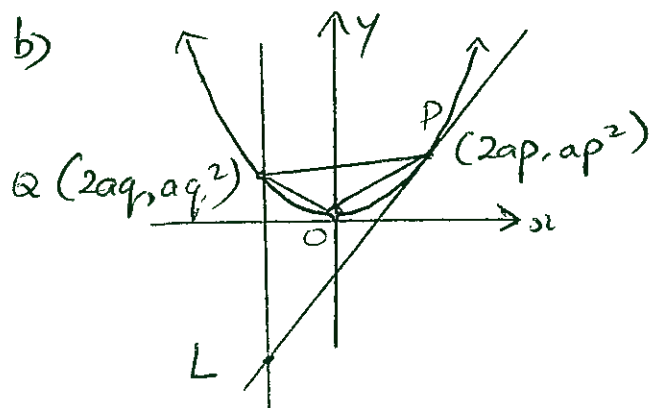


(ii) Need A. solve simult.

$$\begin{aligned} y &= 2x + 1 \quad \text{and} \quad y = 2 - x \\ 2x + 1 &= 2 - x \\ 3x &= 1 \\ x &= \frac{1}{3} \end{aligned} \quad (1)$$

$\therefore$ solution to $|x - 2| < 2x + 1$

$$x > \frac{1}{3} \quad (1)$$



(i) $m_{OP} = \frac{ap^2 - 0}{2ap - 0}$

$$= \frac{p}{2} \quad (1)$$

(ii) $m_{OQ} = \frac{q}{2}$ lines $\perp \therefore (1)$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4 \quad (1)$$

(iii) $x^2 = 4ay \rightarrow y = \frac{x^2}{4a}$

$$y' = \frac{2x}{4a}$$

at $x = 2ap$ $y' = \frac{4ap}{4a}$

$$= p \quad (1)$$

$\therefore$ tangent $y - ap^2 = p(x - 2ap)$

$$y = px - ap^2 \quad (1)$$

(iv) at L $x = 2aq \therefore$

$$y = p(2aq) - ap^2$$

$$= 2apq - ap^2 \quad (1)$$

$\therefore L (2aq, 2apq - ap^2)$

$$pq = -4$$

$$\therefore q = -\frac{4}{p}$$

$$\therefore x = 2a\left(-\frac{4}{p}\right)$$

$$x = -\frac{8a}{p} \Rightarrow p = \frac{-8a}{x} \quad (1)$$

$$y = 2apq - ap^2$$

$$y = 2a\left(-\frac{4}{p}\right) - a\left(\frac{-8a}{x}\right)^2$$

$$y = -8a - \frac{64a^3}{x^2}$$

$$x^2 y = -8ax^2 - 64a^3 \quad (1)$$

$$8ax^2 + x^2 y + 64a^3 = 0$$

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Question 6

a) $\int_0^{\frac{2\pi}{3}} \sec^2 x \tan^2 x \, dx$

$$\frac{d}{dx} (\tan^3 x) = 3 \sec^2 x \tan^2 x \quad (1)$$

$$\therefore \int_0^{\frac{2\pi}{3}} \sec^2 x \tan^2 x \, dx = \frac{1}{3} (\tan^3 x) \Big|_0^{\frac{2\pi}{3}} \quad (1)$$

$$= \frac{1}{3} \left[\tan^3 \frac{2\pi}{3} - \tan^3 0 \right]$$

$$= \frac{1}{3} \cdot (-3\sqrt{3})$$

$$= -\sqrt{3} \quad (1)$$

b) $\log_a 3 = x \quad \log_a 4 = y$

$$\log_3 6 = \frac{\log_a 6}{\log_a 3} \quad (1)$$

$$= \frac{\log_3 3 \times \sqrt{4}}{\log_a 3}$$

$$\begin{aligned}
 b) &= \frac{\log_a 3 + \frac{1}{2} \log_a 4}{\log_a 3} \\
 &= \frac{x + \frac{1}{2}y}{x} \quad \left. \vphantom{\frac{x + \frac{1}{2}y}{x}} \right\} \textcircled{1} \\
 &= 1 + \frac{y}{2x}
 \end{aligned}$$

c) (i) 2000×1.05^n $\textcircled{1}$

(ii) Investments =

$$\begin{aligned}
 &2000 \times 1.05^n + 2000 \times 1.05^{n-1} + \dots \\
 &\quad \dots + 2000 \times 1.05.
 \end{aligned}
 \quad \left. \vphantom{2000 \times 1.05^n + 2000 \times 1.05^{n-1} + \dots} \right\} \textcircled{1}$$

$$\begin{aligned}
 S_n &= 2000 \times 1.05 \left(\frac{1.05^n - 1}{1.05 - 1} \right) \\
 &= \underline{42000(1.05^n - 1)} \quad \textcircled{1}
 \end{aligned}$$

(iii) $93454.20 = 42000(1.05^n - 1)$

$$1.05^n = \frac{93454.2}{42000} + 1.$$

$$1.05^n = 3.2251$$

$$\begin{aligned}
 n &= \frac{\ln(3.2251)}{\ln(1.05)} \\
 n &= \underline{24} \quad \left. \vphantom{n = \frac{\ln(3.2251)}{\ln(1.05)}} \right\} \textcircled{1}
 \end{aligned}$$

d) $y = ax^3 - 7x^2 + bx + 20$

(2,0) satisfies.

$$0 = a(2)^3 - 7(2)^2 + 2b + 20$$

$$0 = 8a - 28 + 2b + 20$$

$$8a + 2b = 8$$

$$4a + b = 4 \quad \text{---} \textcircled{A} \quad \textcircled{1}$$

$$\begin{aligned}
 y' &= 3ax^2 - 14x + b \\
 \text{when } x=2 \quad y' &= 0
 \end{aligned}$$

$$\therefore 0 = 3a(2)^2 - 14(2) + b$$

$$0 = 12a - 28 + b \quad \textcircled{1}$$

$$12a + b = 28 \quad \text{---} \textcircled{B}$$

$$4a + b = 4 \quad \text{---} \textcircled{A}$$

$$\textcircled{B} - \textcircled{A} \quad 8a = 24$$

$$a = 3$$

$$b = -8$$

$$\left. \vphantom{a = 3} \right\} \textcircled{1}$$

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Question 7.

$$a) \frac{\sin^3 a + \cos^3 a}{\sin^2 a - \cos^2 a} = \frac{\csc a - \cot a}{1 - \cot a}$$

LHS

$$\begin{aligned}
 &(\sin a + \cos a)(\sin^2 a - \sin a \cos a + \cos^2 a) \\
 &= (\sin a + \cos a)(\sin a - \cos a) \quad \textcircled{1}
 \end{aligned}$$

$$\frac{1 - \sin a \cos a}{\sin a - \cos a}$$

$$\sin^2 a + \cos^2 a = 1 \quad \textcircled{1}$$

$$= \frac{1}{\sin a} - \frac{\sin a \cos a}{\sin a}$$

$$\text{---} \textcircled{1}$$

$$\frac{\sin a}{\sin a} - \frac{\cos a}{\sin a}$$

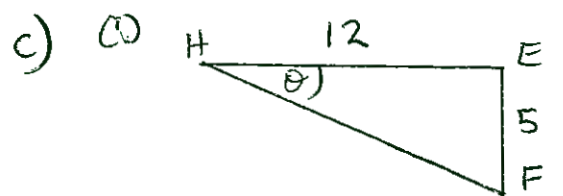
$$= \frac{\csc a - \cot a}{1 - \cot a}$$

= RHS.

$$b(i) \frac{9^x + 6^x}{15^x + 10^x} = a^x$$

$$\text{LHS. } \frac{3^x (3^x + 2^x)}{5^x (3^x + 2^x)} \quad (1)$$

$$\left. \begin{array}{l} \text{RHS} = 0 \\ \frac{3^x + 3^x}{5^x + 5^x} = \left(\frac{3}{5}\right)^x \end{array} \right\} \text{Let } x = \frac{3}{5} \checkmark$$



$$\tan \theta = \frac{5}{12}$$

$$\theta = 22^\circ 37' \quad (1)$$

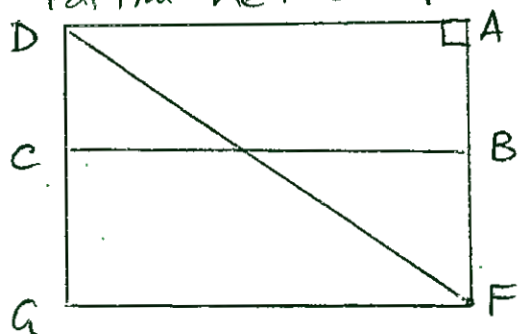
$$EX^2 = 6^2 + 12^2 - 2 \cdot 6 \cdot 12 \cdot \cos 22^\circ 37'$$

$$EX = 6.861 \quad (1)$$

$$\tan \angle AXE = \frac{6}{6.861}$$

$$\therefore \angle AXE = 41^\circ 10' \quad (1)$$

(ii) Partial net of prism



Short distance using Pythagoras

$$DF = \sqrt{12^2 + 11^2}$$

$$= \sqrt{265} \quad (1)$$

$$d(i) y = e^{\frac{1}{2}x} \quad (1)$$

$$(ii) y' = \frac{1}{2} e^{\frac{x}{2}}$$

$$y'' = \frac{1}{4} e^{\frac{x}{2}}$$

$$y''' = \frac{1}{8} e^{\frac{x}{2}} \text{ etc}$$

$$\therefore \text{by } \left(\frac{1}{2} e^{\frac{x}{2}} + \frac{1}{4} e^{\frac{x}{2}} + \dots \right) \quad (1)$$

$$= e^{\frac{x}{2}} \left[\frac{\frac{1}{2}}{1 - \frac{1}{2}} \right]$$

$$= e^{\frac{x}{2}} \cdot 1$$

$$= e^{\frac{x}{2}} \quad (1)$$

$$= y.$$

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trivial case
 $\frac{1}{2}y$ $y=0$... must
 Qn

$$\frac{1}{2}y + \frac{1}{2} \times \frac{1}{2}y + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}y + \dots$$